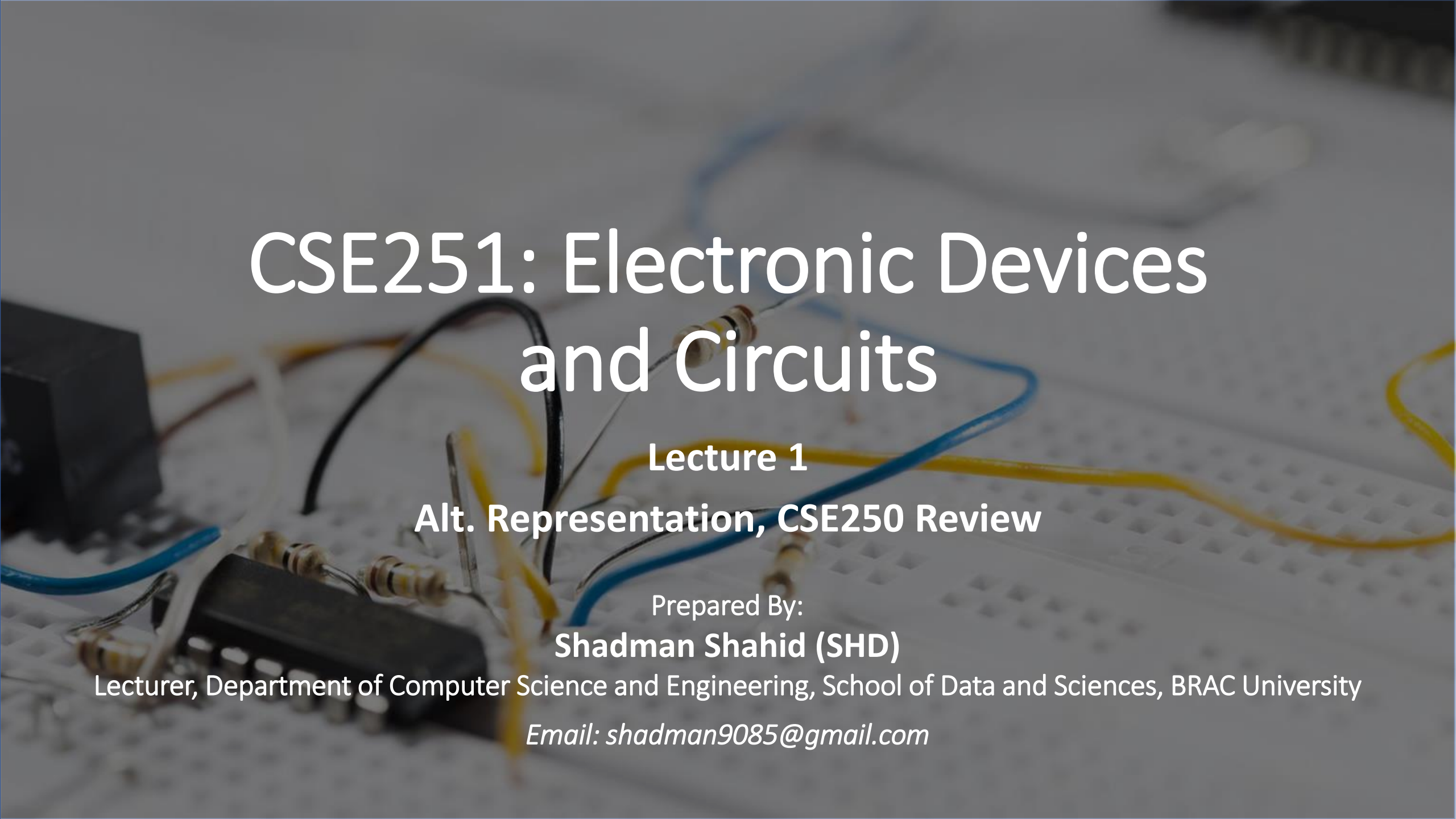




CSE251: Electronic Devices and Circuits

Lecture 1

Alt. Representation, CSE250 Review

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Outline

- Alternative Circuit Representation – Line diagrams
- CSE250 Review
 - KCL, KVL
 - Series, Parallel resistor network – Voltage Division, Current division
 - Examples
- IV Characteristics
 - Linear IV – Resistors, Voltage Source, Current Source, SC, OC.
 - Non-Linear IV – Piecewise Linear Model

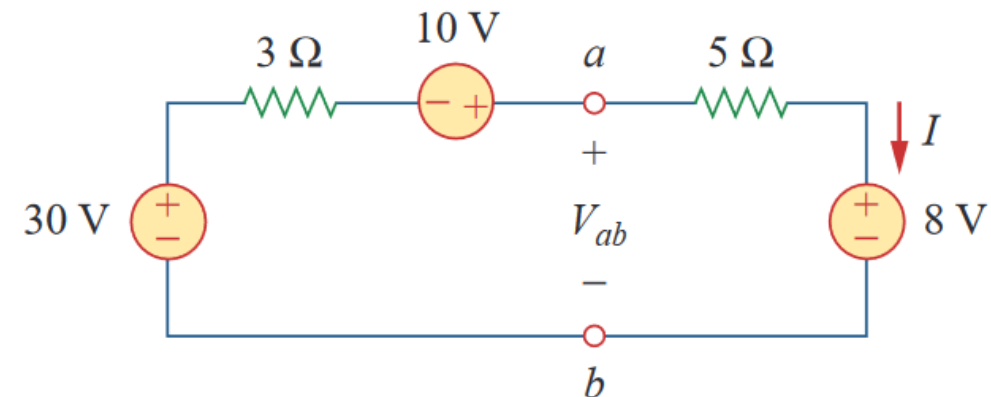
Alternative Circuit Representation: Line diagrams

Steps to decompose circuits to line diagram

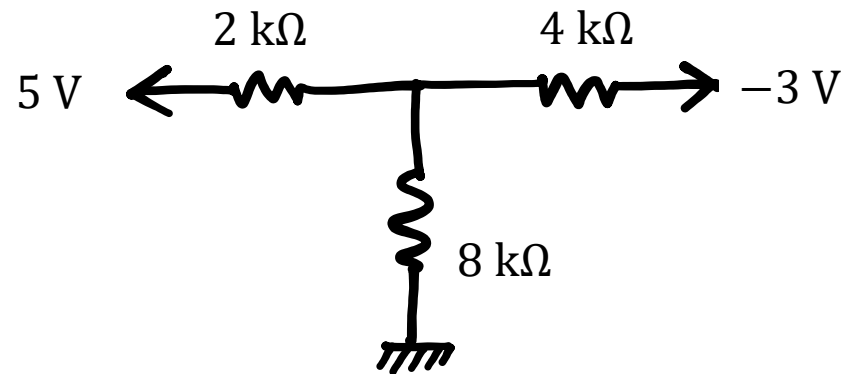
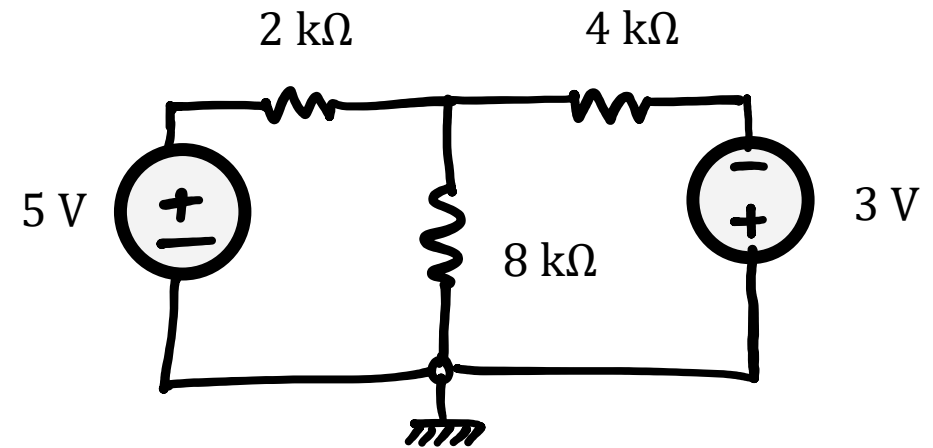
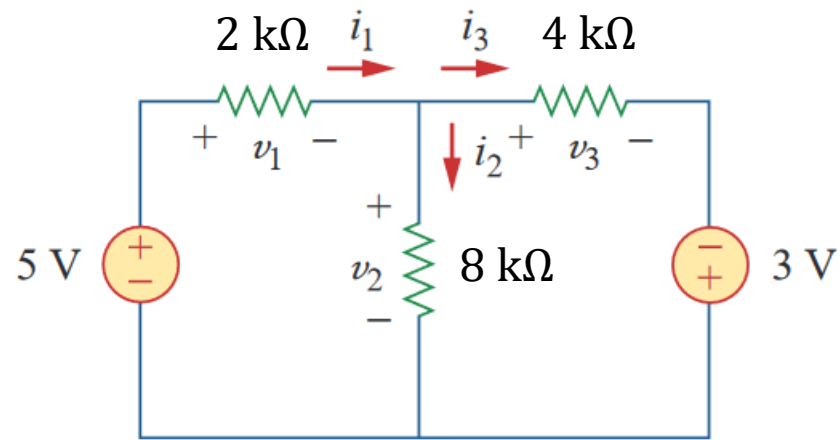
1. Set a ground so that number of **floating voltage** sources are minimized.
2. Detach the ground **from the voltage source**.
3. Convert the non-floating voltage sources (~~current sources~~) into:
 - Arrow : ($\rightarrow$) **Fixed/Constant voltage source**
4. Keep passive elements as they are.

Floating voltage sources:

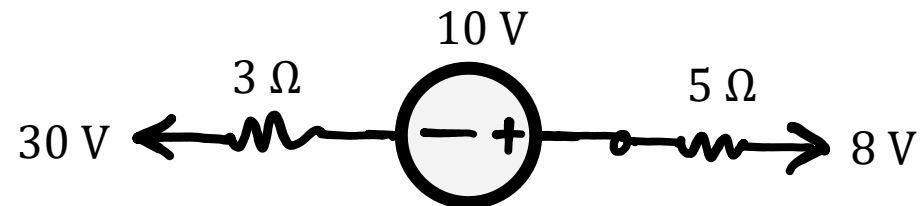
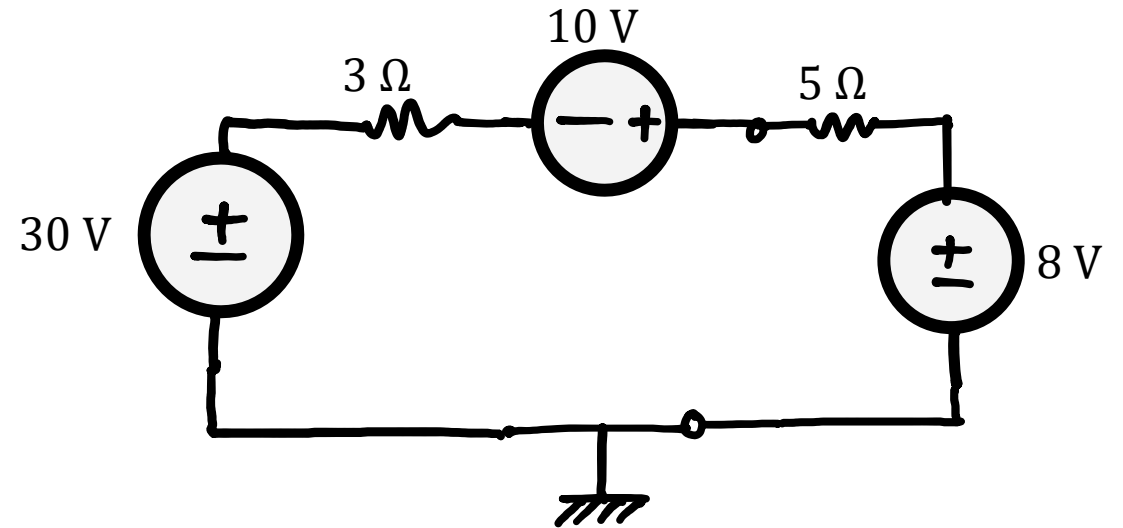
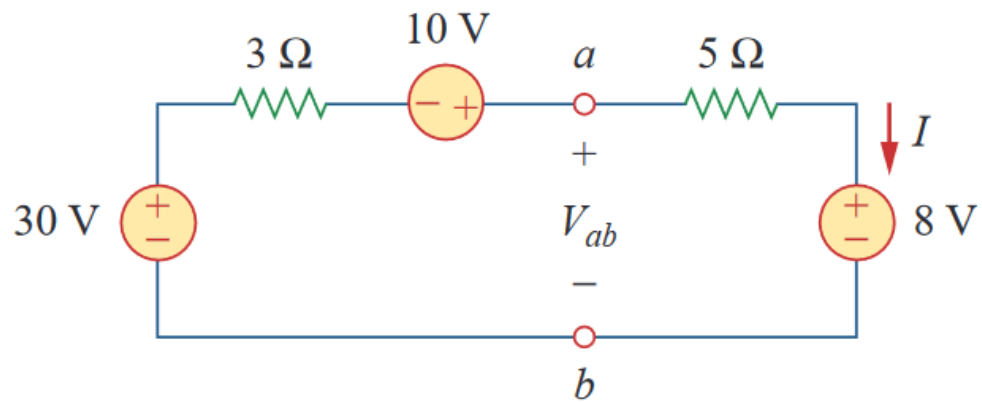
Voltage sources which are **not connected the ground** terminal. In the diagram, the **10 V** voltage source is floating



Line diagrams: Example 1

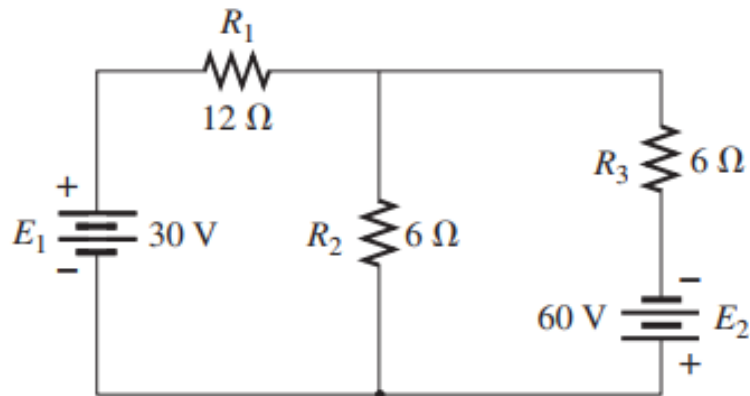


Line diagrams: Example 2



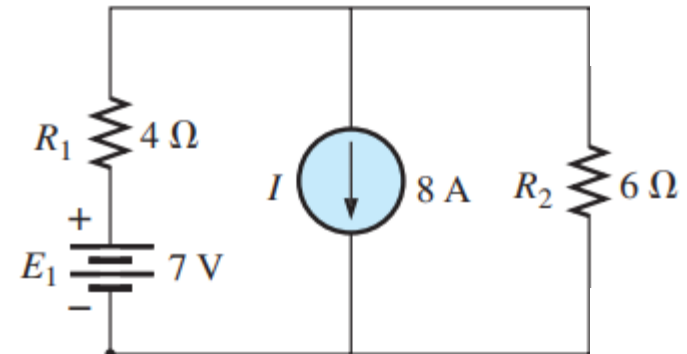
More Examples

Difficulty : 2/5



Example: 2

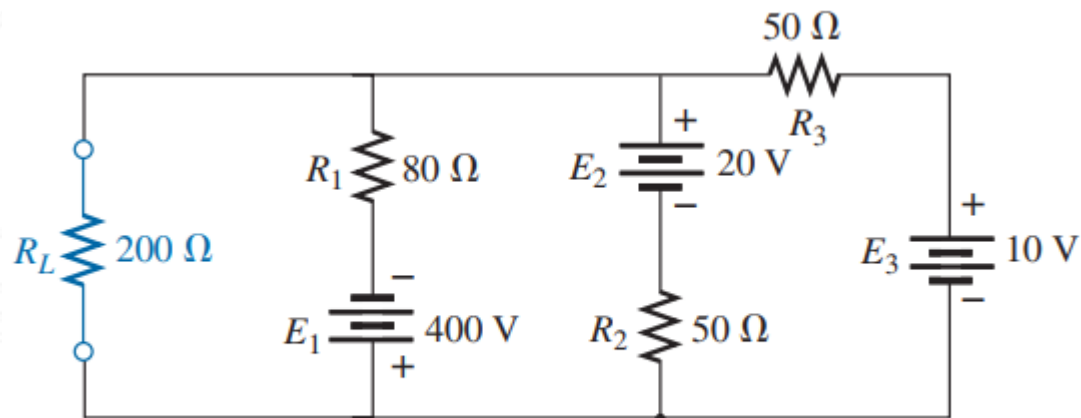
Difficulty : 3/5



Example: 3

More Examples

Difficulty : 4/5



Example: 4

Step – (4) Make all the active elements (dc/ac type, voltage/current sources) into single terminals (arrows/circles) using the voltages you wrote as much as you can **[THERE MIGHT BE CASES WHERE YOU CAN'T DO THAT]**

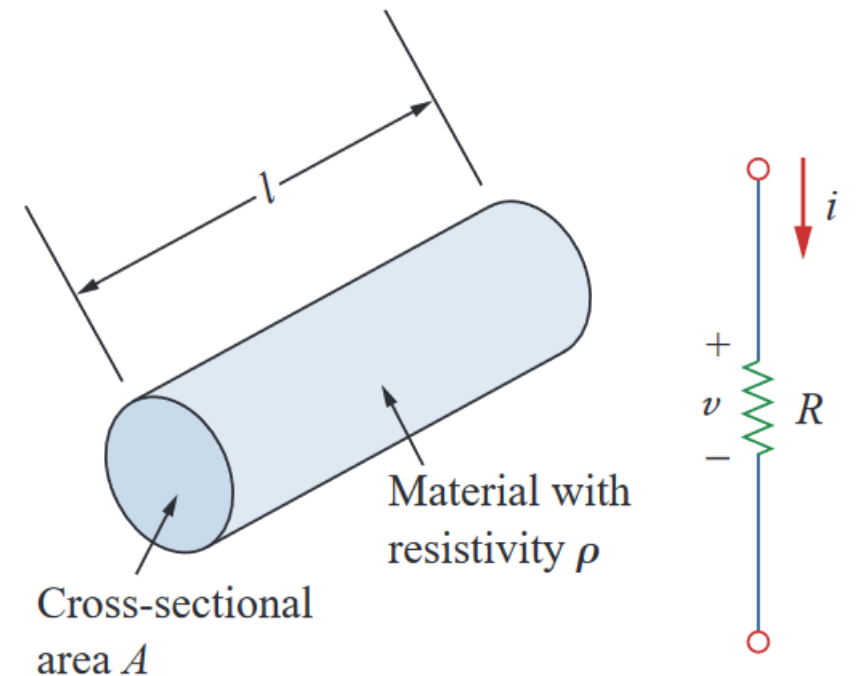
The fundamentals ...

Ohm's Law –

- the voltage v across a resistor is **directly proportional** to the current i flowing through the resistor (R)

$$v \propto i$$

$$v = iR$$



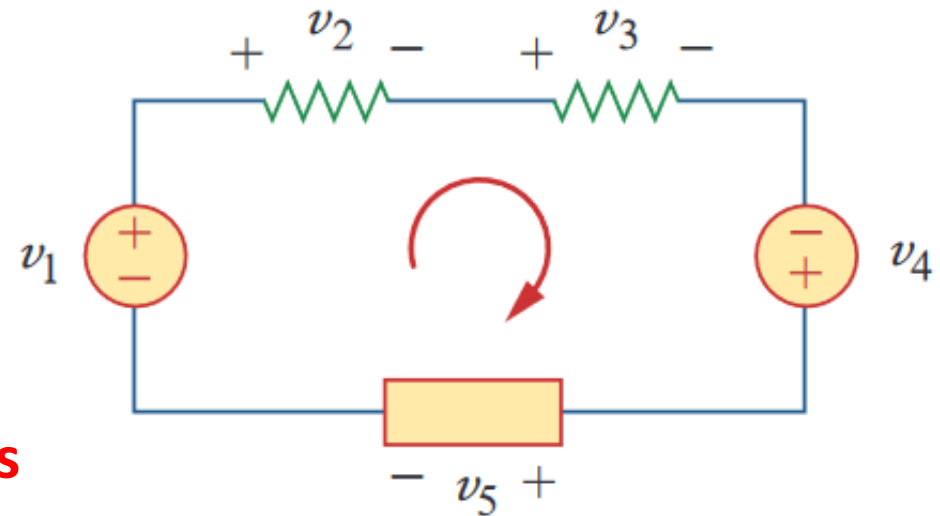
KVL: Kirchhoff's voltage law

The algebraic sum of all **voltages** around **a closed path (or loop)** is zero.

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

$$v_2 + v_3 + v_5 = v_1 + v_4$$

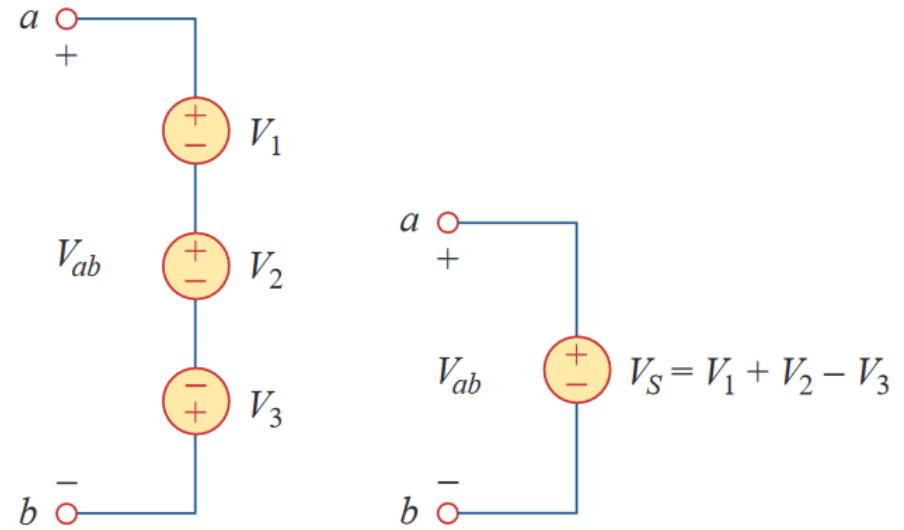
Sum of voltage drops = **Sum of voltage rises**



KVL: Kirchhoff's voltage law

$$-V_{ab} + V_1 + V_2 - V_3 = 0$$

$$V_{ab} = V_1 + V_2 - V_3$$



Equivalent Circuits

KVL – Example 1

Find I and V_{ab} in the circuit

Solution:

KVL

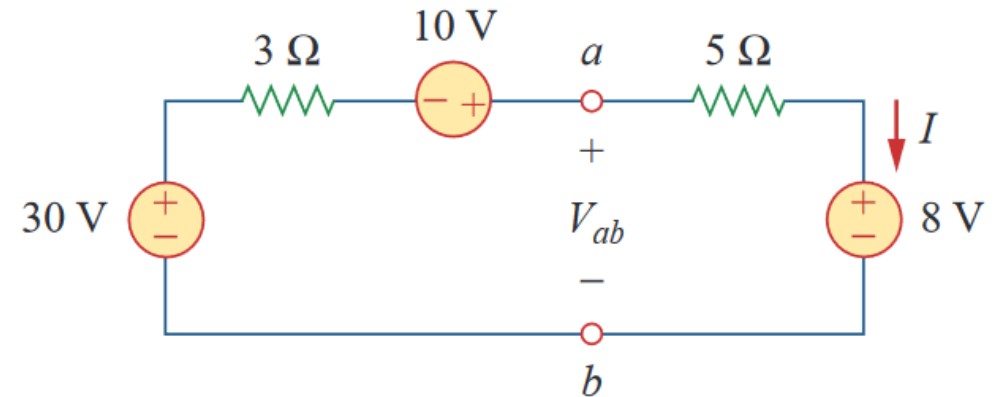
$$-30 + 3I - 10 + 5I + 8 = 0$$

$$I = \frac{32}{8} \text{ A} = 4 \text{ A}$$

KVL

$$-V_{ab} + 5I + 8 = 0$$

$$V_{ab} = 28 \text{ V}$$



Tip: If you find resistance values in $\mathbf{k\Omega}$ instead of $\mathbf{\Omega}$, don't convert the $\mathbf{k\Omega}$ values to $\mathbf{\Omega}$. Just find currents in $\mathbf{mA}$ instead of $\mathbf{A}$.

KVL – Example 2

Find v_1, v_2, v_3, i_1, i_2 and i_3 in the circuit

Solution:

KVL in first loop

$$-5 + 2i_1 + 8(i_1 - i_3) = 0$$

$$10i_1 - 8i_3 = 5$$

KVL in second loop

$$-8(i_1 - i_3) + 4i_3 - 3 = 0$$

$$-8i_1 + 12i_3 = 3$$

Solving:

$$i_1 = 1.5 \text{ mA}$$

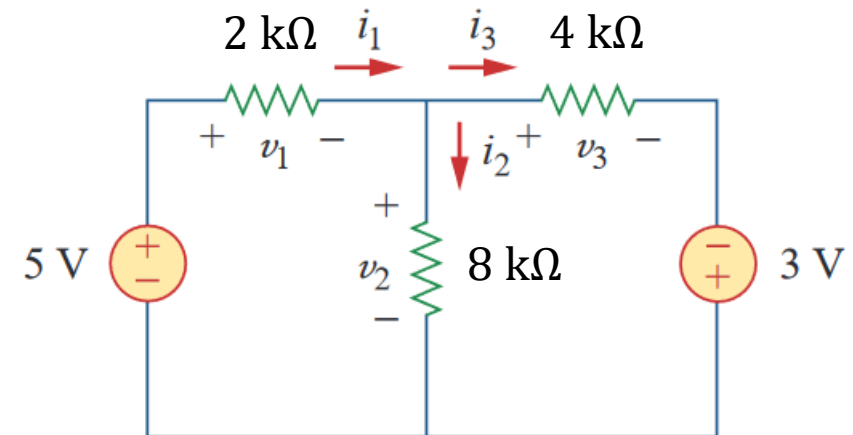
$$i_3 = 1.25 \text{ mA}$$

$$i_2 = i_1 - i_3 = 0.25 \text{ mA}$$

$$v_1 = 3 \text{ V}$$

$$v_2 = 2 \text{ V}$$

$$v_3 = 5 \text{ V}$$



Tip: If you find resistance values in **kΩ** instead of **Ω**, don't convert the **kΩ** values to **Ω**. Just find currents in **mA** instead of **A**.

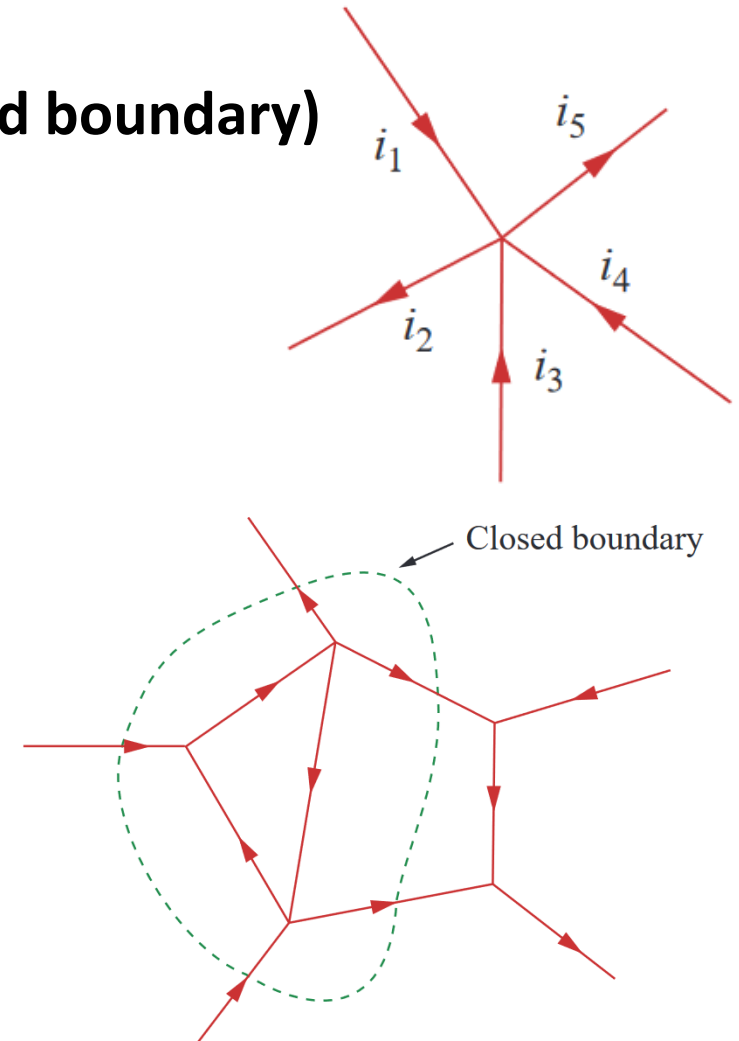
KCL: Kirchhoff's Current Law

The algebraic sum of the **currents** entering a **node (closed boundary)** is equal to the sum of the currents leaving the node.

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

Current Entering node: **Positive**
Current Exiting node: **Negative**

Or vice versa...



KCL- Example 1

Find v_1, v_2, v_3, i_1, i_2 and i_3 in the circuit

Solution:

KCL in node v_a . (PS: $v_a = v_2$)

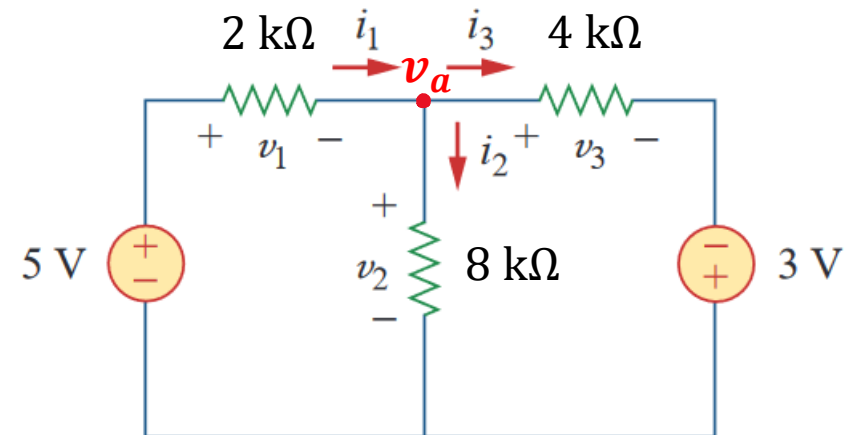
$$\frac{5 - v_2}{2} - \frac{v_2 - (-3)}{4} - \frac{v_2 - 0}{8} = 0$$

$$v_2 \left(-\frac{1}{2} - \frac{1}{4} - \frac{1}{8} \right) = -\left(\frac{5}{2} - \frac{3}{4} \right)$$

$$v_2 = \frac{7}{4} \cdot \frac{8}{7} \text{ V} = 2 \text{ V}$$

$$v_1 = 5 - v_2 = 3 \text{ V}$$

$$v_3 = v_2 - (-3) = 5 \text{ V}$$

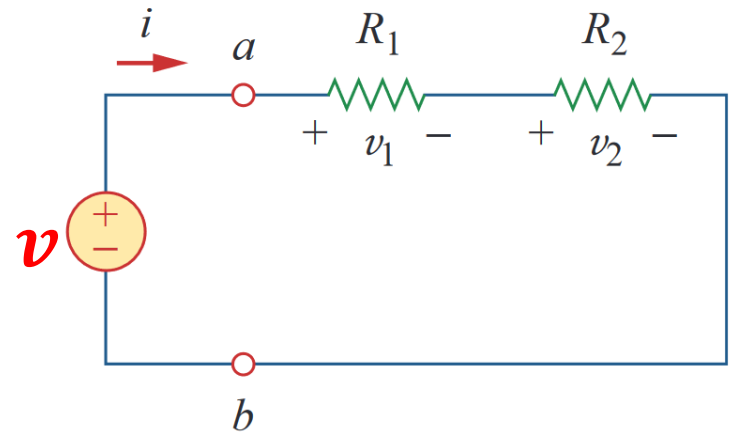


Series Resistors and Voltage Division

The **equivalent resistance** of any number of resistors connected in **series** is the sum of the individual resistances.

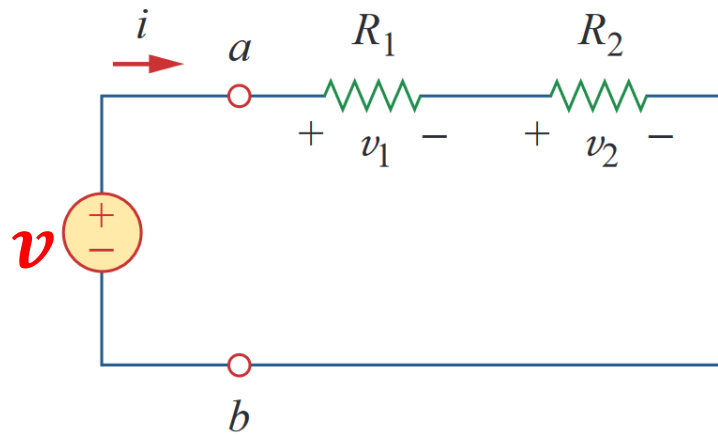
Principle of voltage division

Source voltage v - is divided among the resistors in direct proportion to their resistances; the larger the resistance, the larger the voltage drop.



$$v_1 = \frac{R_1}{R_1 + R_2} v \quad v_2 = \frac{R_2}{R_1 + R_2} v$$

Line diagram: Example 3



$$v_2 = \frac{R_2}{R_1 + R_2} v$$

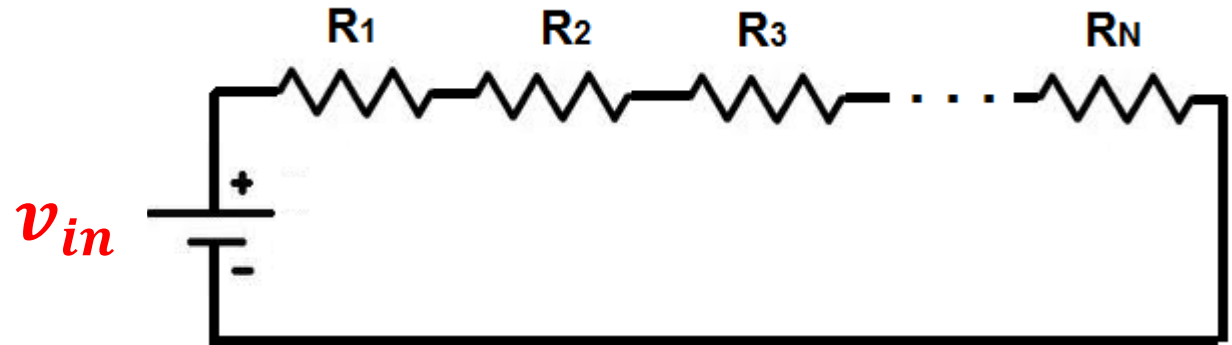
A diagram illustrating the voltage divider principle. It shows two resistors, R_1 and R_2 , connected in series. The total voltage v is applied across the series combination. The voltage v_2 is the voltage across resistor R_2 . The diagram shows R_1 at the top and R_2 at the bottom, with a ground symbol at the bottom. Brackets indicate the voltage across each resistor: v_1 across R_1 and v_2 across R_2 . The total voltage v is indicated by a red arrow pointing upwards from the ground symbol.

Series Resistors and Voltage Division

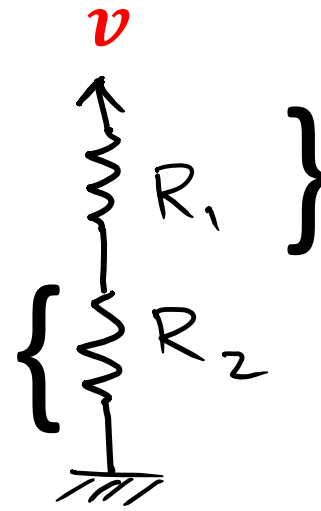
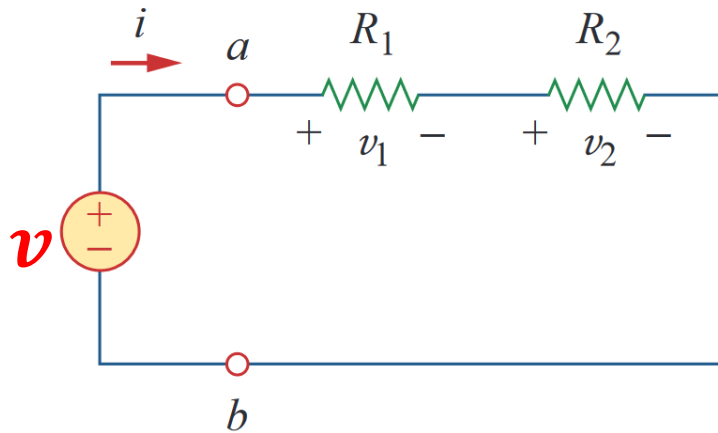
- If there are N resistors in series, the voltage across the i –th resistor is given by,

As $V \propto R$

$$v_i = \frac{R_i}{\sum_i R_i} v_{in}$$



Line diagram: Example 3



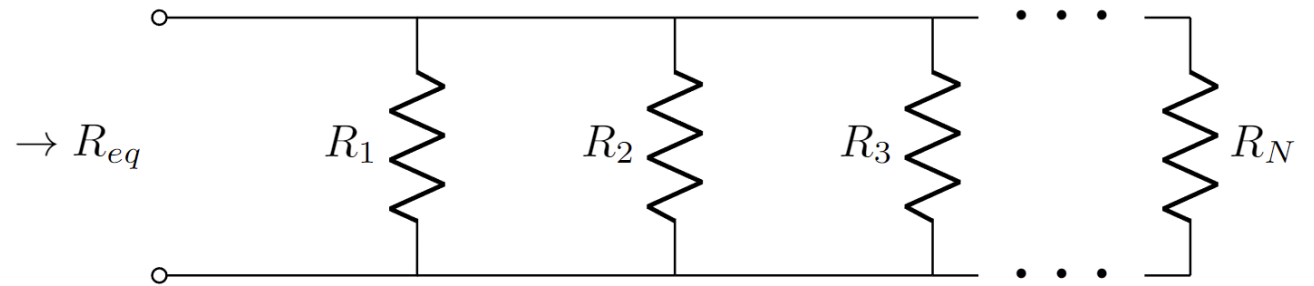
KVL (acts along a line instead of a loop)

$$v - iR_1 - iR_2 = 0$$

Parallel Resistors and Current Division

The **equivalent resistance** of any number of resistors connected in **parallel** is the inverse of the sum of the individual conductances.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots \frac{1}{R_N}$$



Simplification for the case when $R_1 = R_2 = R_3 \cdots = R_N$

$$R_{eq} = \frac{R_1}{N}$$

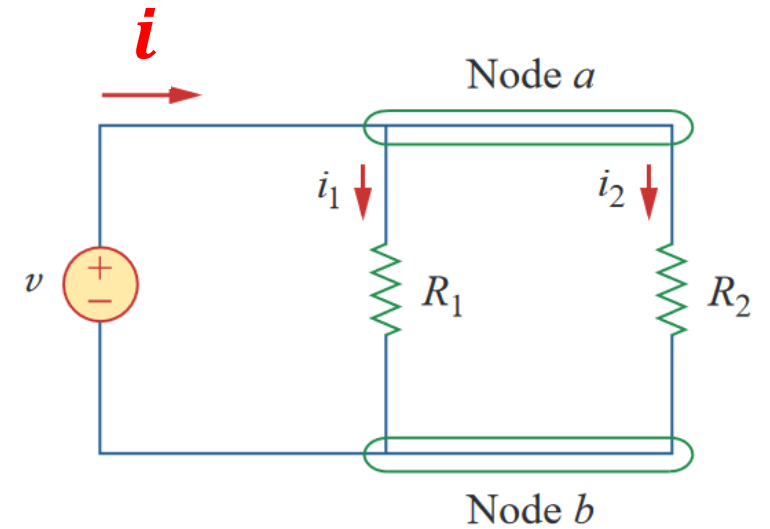
Parallel Resistors and Current Division

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$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad R_{eq} = R_1 || R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

Simplification for the case when $R_1 = R_2$

$$R_{eq} = \frac{R_1}{2}$$



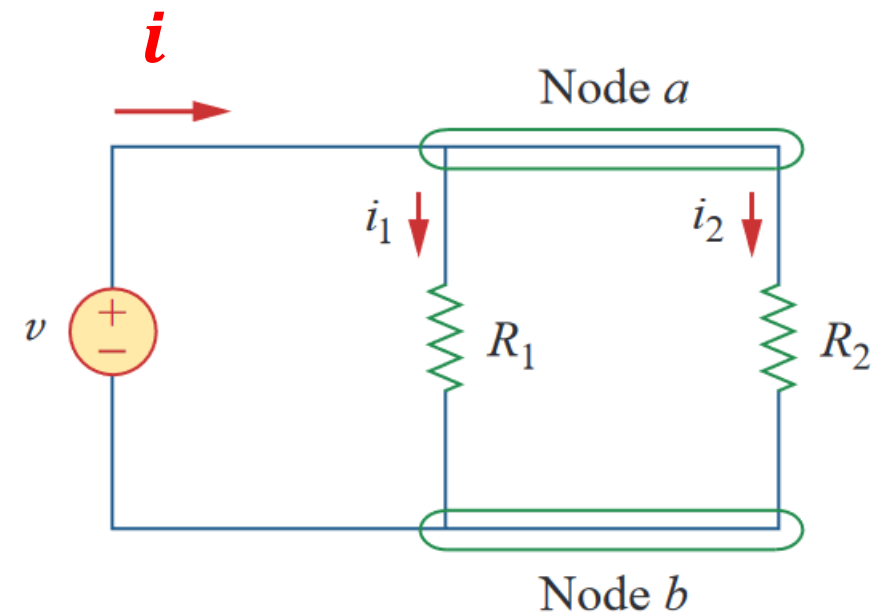
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The **equivalent resistance** of any number of resistors connected in **parallel** is the inverse of the sum of the individual conductances.

Principle of current division

Source current i - is divided among the resistors in direct **inverse** proportion to their resistances; the larger the resistance, the larger the voltage drop.

$$i_1 = \frac{1/R_1}{1/R_1 + 1/R_2} i \quad i_2 = \frac{1/R_2}{1/R_1 + 1/R_2} i$$

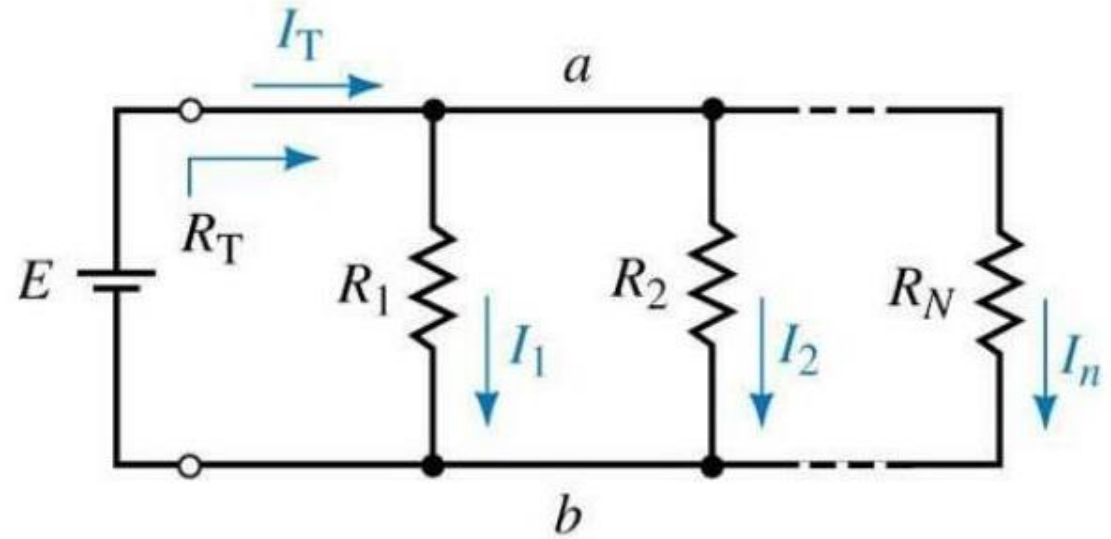


Parallel Resistors and Current Division

- If there are N resistors in parallel, the current through the i -th resistor is given by, $i \in \{1, 2, 3, \dots, N\}$

$$\text{As } I \propto \frac{1}{R}$$

$$I_i = \frac{1/R_i}{\sum_i 1/R_i} I_T$$



Thank You